

VARNAJA UTHAYARAJ

AI/ML | Software Engineering | Iot & Networking

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in Varnaja Uthayaraj

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🐙 VeeQubit

PROFESSIONAL SUMMARY

As a computer engineering student, I am passionate about Software Engineering, AI, ML and Networking. And I have built a strong theoretical foundation complemented by practical skills, honed through coursework and hands-on projects. I am eager to tackle real world challenges, contribute to innovative solutions, and continuously develop my expertise in a dynamic and rewarding environment.

EDUCATION

2023 - present	BSc Eng at University of Jaffna	(CGPA: 3.07/4.0)
2018 - 2022	A/L s at V/Rambaikulam G.M.V	(A2B)

RESEARCH PROJECT

Sentiment and Intent Detection in Code-Mixed Text Dec 2025 - present

- Developed a MuRIL-based multi-task Transformer for Tamil-English code-mixed sentiment, intent, and sarcasm detection.
- Built a gold-standard dataset through multi-annotator labeling and inter-annotator agreement, and integrated expressive linguistic features through feature fusion for sarcasm detection.

PROJECTS

Equipment Request Management System [Github](#)

Developed a web-based Equipment Request Management System using React, Spring Boot, MySQL, enabling equipment request submission, approval workflows, request tracking, and inventory management. As a Project Manager led a cross-functional team of [5] members through Agile Scrum, orchestrating sprint planning, task allocation, and stakeholder communication to deliver the Equipment Request Management System on schedule.

Artisan Gallery-Full Stack Inventory Management System [Github](#)

Developed Artisan Gallery, a full-stack inventory management web application using React.js, Tailwind CSS, Node.js, Express.js, MySQL, and JWT, enabling secure administrator authentication, product and category management, inventory tracking, responsive dashboard analytics, and complete CRUD operations through RESTful APIs. Designed and implemented a modern responsive admin interface with reusable components, database integration, and protected routes to support efficient inventory management workflows.

Me Plus - Multi-Client Task Management Platform for Freelancers [Github](#)

Designed and developed a lightweight, freelancer workspace using React, TypeScript, Tailwind CSS, Node.js, and Express. Applied user-centred design and HCI principles to create an intuitive, responsive interface with client/project management, interactive Kanban/Calendar/List views, task tracking, live time logging, deadline monitoring, invoicing, and AI-assisted productivity features. Focused on usability, cognitive ergonomics, efficient interactions, and consistent UI/UX design.

Real-time Video Conferencing Application - Linkzo Live

[Github](#)

Developed Linkzo live, a real-time video conferencing web application using WebRTC, JavaScript, Node.js, Express, and Socket.IO, enabling peer-to-peer video/audio communication, text chat, screen sharing, room-based conferencing, and media controls including mute/unmute and camera toggle. Collaborated in a team to design, implement, and test real-time communication features for seamless user interaction.

Dam Safety Monitoring System

[Github](#)

Engineered an IoT Smart Dam Monitoring system with ESP32 sensors and a Node.js backend, delivering real-time water level data, remote gate control, and predictive analytics that reduced manual inspection time by [95%]. Contributed to web application development and embedded system integration within a collaborative team environment.

Electronic Waste Detection and Classification System

[Github](#)

Developed a deep learning computer vision system using YOLO26n to detect and classify 37 categories of electronic waste from images and real-time browser camera input; curated and validated a 7,210-image dataset, converting annotations to YOLO format with train/validation/test splits, and trained the model via transfer learning, achieving 92.3% precision, 86.3% recall, 91.9% mAP@50, and 75.6% mAP@50-95 on an independent test set, while building a Streamlit evaluation application to visualize predictions, ground-truth annotations, confidence scores, IoU, and TP/FP/FN metrics. Skills: Python, YOLO, PyTorch, Computer Vision, Streamlit.

VOLUNTEERING

Vice Media and Content Coordinator

Oct 2025 - present

IEEE Communications Society (ComSoc) Student Chapter - University of Jaffna

Editor

2025 - present

Computer Engineering Society (CES)- University of Jaffna

SKILLS

Programming Languages	Python, Java, C++, JavaScript
Frontend Development	React, HTML, CSS
Backend Development	Spring Boot, Flask, Node.js, Express.js
Databases	MySQL, MongoDB
Frameworks & Libraries	Streamlit, NumPy, Pandas, Scikit-learn, OpenCV, LangChain
Tools & Platforms	Git, GitHub, Jupyter Notebook, Postman, JIRA, VS Code

REFERENCES

Dr. (Mrs.) J. Jarachanthan

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